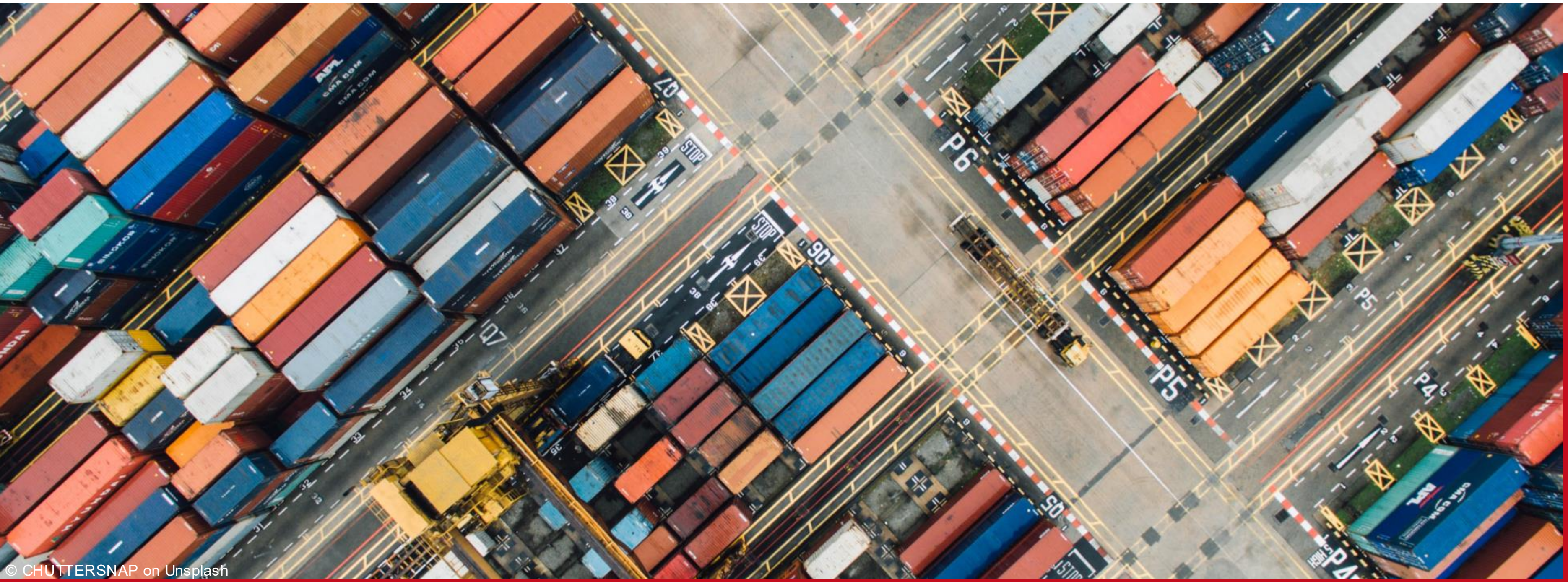
TECHNISCHE UNIVERSITÄT BERLIN



StreamShard: A Distributed Event Aggregation Engine

Jakob Al Khuzayi & Javier Herrero Delgado | Scalability Engineering | July 13, 2026

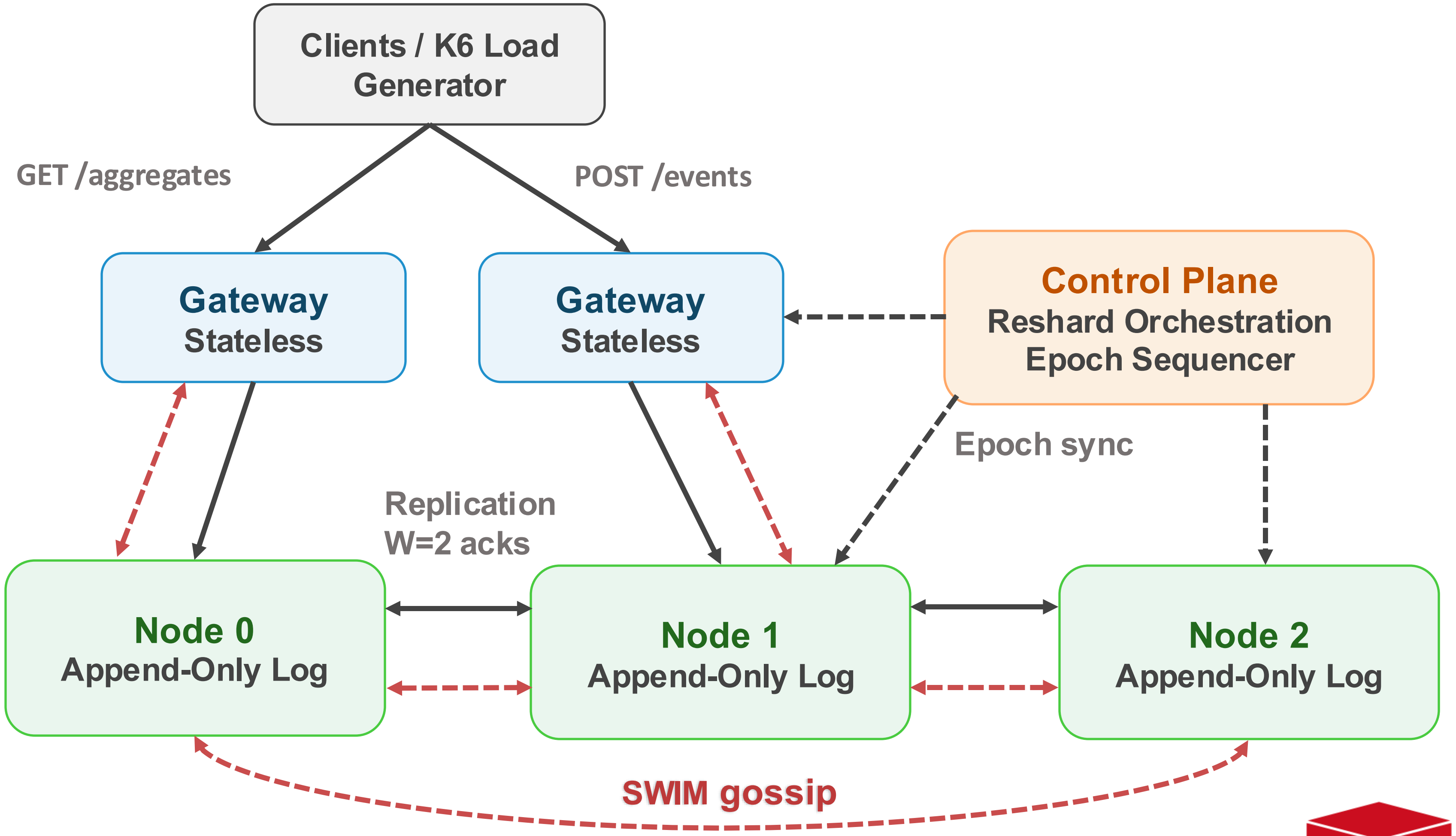
https://github.com/Jakob-al28/StreamShard

Architecture

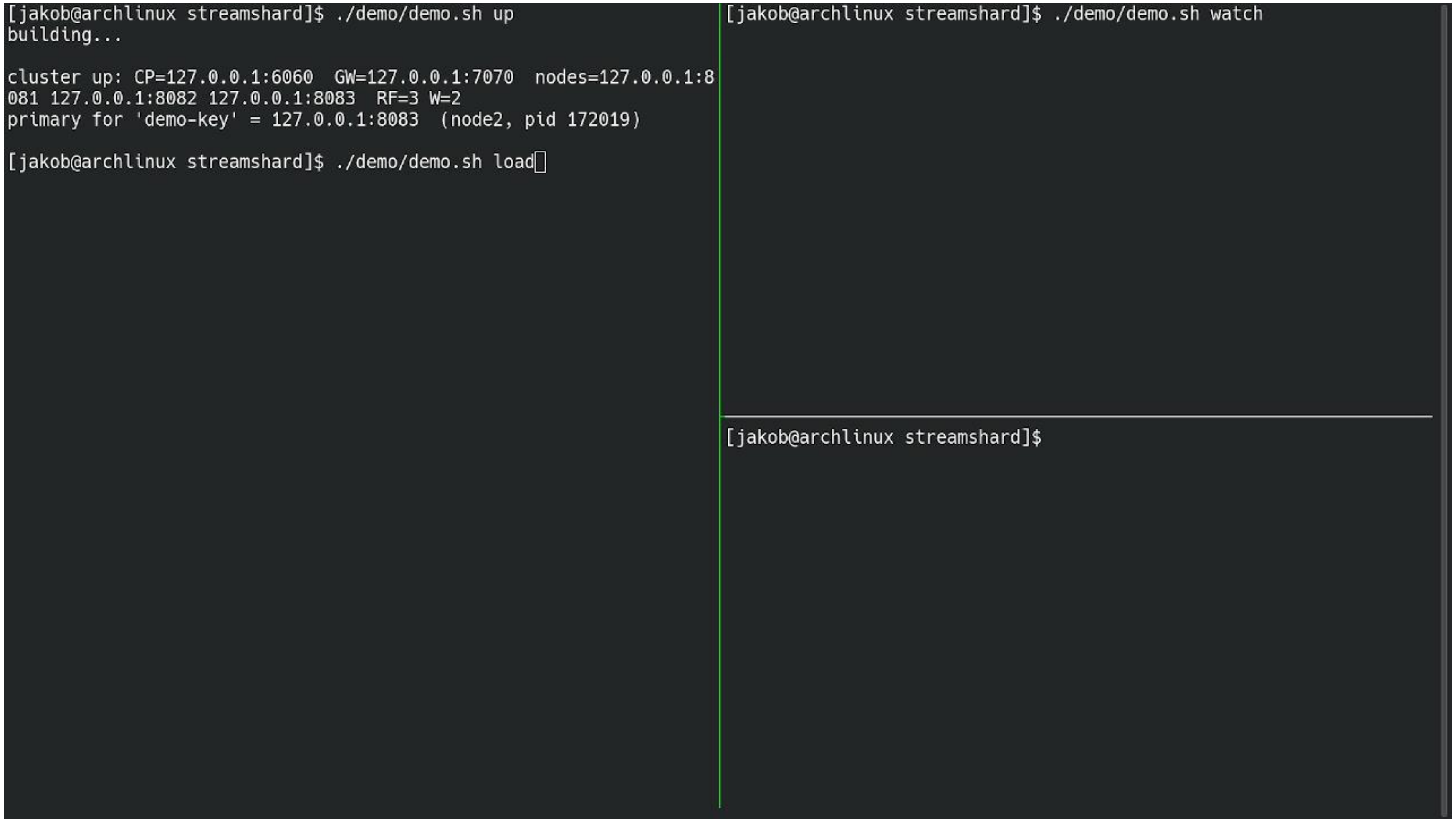
SWIM gossip
Membership Management

Gateway
Token-Bucket
Circuit Breaker
Consistent Hash Ring
Write Quorum Replication

Data Nodes
Load Shedding
Rolling Aggregates
Compaction Snapshots
Epoch Fencing



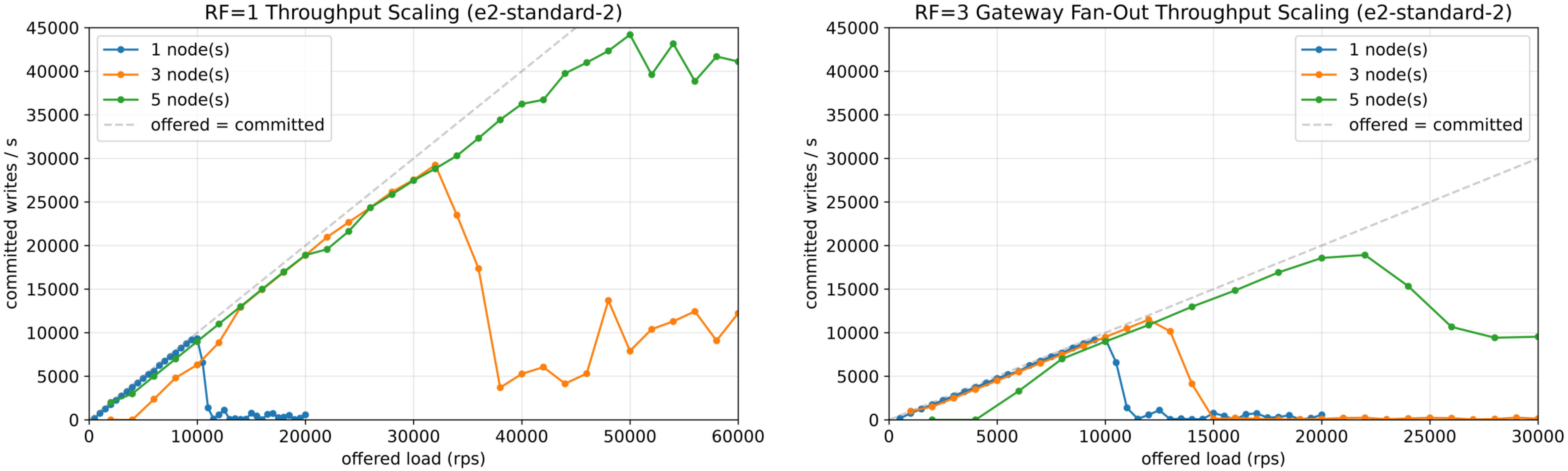
Demo: Primary Failover and Live Resharding



 Jakob Al Khuzayi & Javier Herrero Delgado | Scalability Engineering

Scalability Results: The Cost of Replication

Write Quorum replication (RF=3, W=2) reduces peak throughput by **~60%**

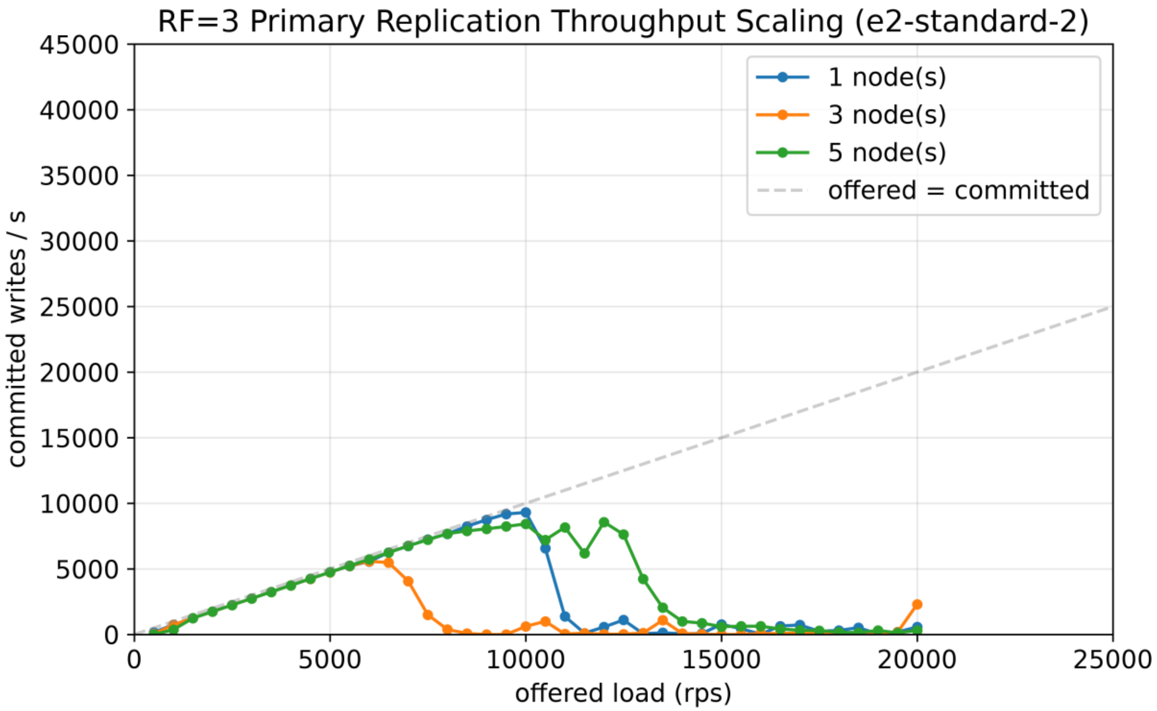


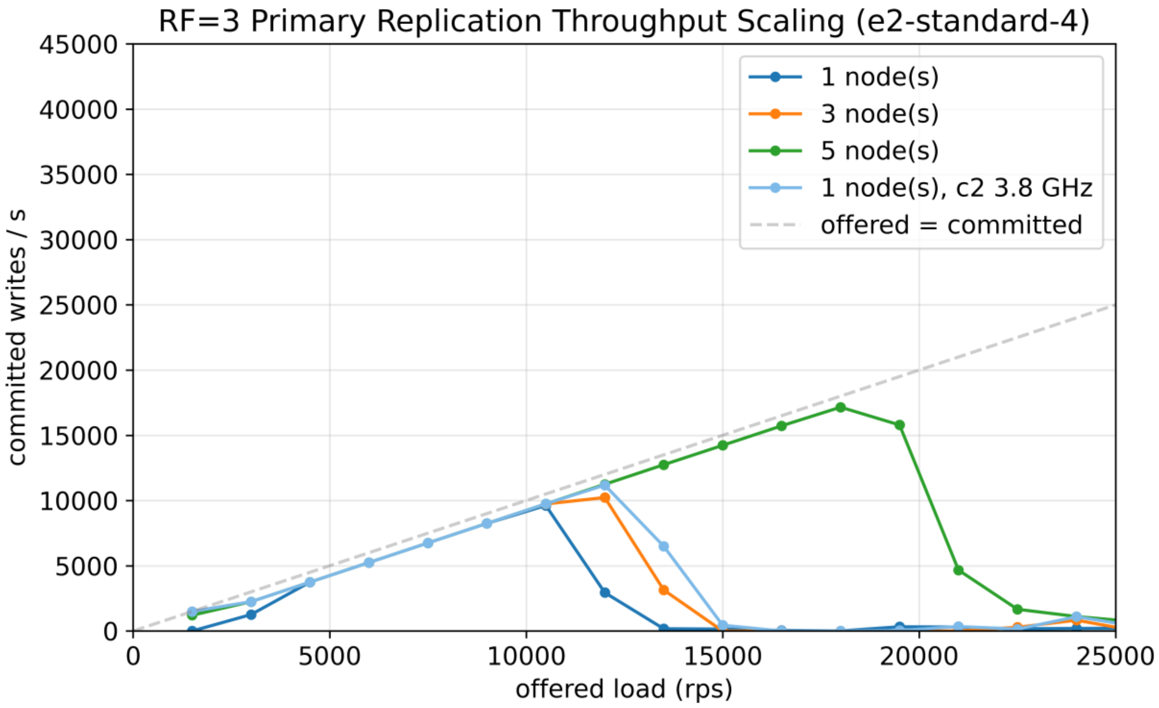
 Jakob Al Khuzayi & Javier Herrero Delgado | Scalability Engineering

Scalability Results: Leader-Based Replication & Vertical Scaling

Primary-driven replication costs **~55%** throughput vs. gateway fan-out

Core count helps parallel work, clock speed helps serial work





System Limitations & Future Work

Per-node global lock on the WAL

Serialized commits limit throughput at **~10,000 writes/s** per node

Scale Up & Parallel Write:

Shard partition by key across workers

Centralized Control Plane

Single coordinator manages epochs and resharding

Replicated Control Plane

backed by a consensus store (e.g., ZooKeeper, etcd).

See the README for a full list of known limitations.